main.c

Run ▶

```c
#include <stdio.h>
#include <string.h>

char stack[100], input[100];
int top = -1;

char precedence[4][4] = {
    {'>', '<', '<', '>'},
    {'>', '>', '<', '>'},
    {'>', '>', ' ', '>'},
    {'<', '<', '<', 'A'}
};

char terminal(char c)
{
    if (c == 'i') return 'i';
    if (c == '+' || c == '*') return c;
    return '$';
}

int indexOf(char c)
{
    if (c == '+') return 0;
    if (c == '*') return 1;
    if (c == 'i') return 2;
    return 3;
}
```

Output

```
Enter expression (use i for identifier): i+i

Stack   Input   Action
$   i+i$    Shift
$i  +i$ Reduce E->i
$E  +i$ Shift
$E+ i$  Shift
$E+i    $   Reduce E->i
$E+E    $   Accept
```

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Run ▶

```c

int main()
{
    int i = 0;
    char a, b;

    printf("Enter expression (use i for identifier): ");
    scanf("%99s", input);

    strcat(input, "$");
    stack[++top] = '$';

    printf("\nStack\tInput\tAction\n");

    while (1)
    {
        a = terminal(stack[top]);
        b = terminal(input[i]);

        printf("%s\t%s\t", stack, input + i);

        if (input[i] == '$' && stack[top] == 'E')
        {
            printf("Accept\n");
            break;
        }

        if (input[i] == '$' && stack[top] == '$')
```

Output

```
Enter expression (use i for identifier): i+i

Stack   Input   Action
$   i+i$    Shift
$i  +i$ Reduce E->i
$E  +i$ Shift
$E+ i$  Shift
$E+i    $   Reduce E->i
$E+E    $   Accept
```

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```
55      if (input[i] == '$' && stack[top] == '$')
56      {
57          printf("Reject\n");
58          break;
59      }
60
61      if (input[i] == 'i')
62      {
63          stack[++top] = 'i';
64          stack[top + 1] = '\0';
65          i++;
66          printf("Shift\n");
67      }
68      else if (stack[top] == 'i')
69      {
70          stack[top--] = '\0';
71          stack[++top] = 'E';
72          stack[top + 1] = '\0';
73          printf("Reduce E->i\n");
74      }
75      else if (input[i] == '+' || input[i] == '*')
76      {
77          stack[++top] = input[i++];
78          stack[top + 1] = '\0';
79          printf("Shift\n");
80      }
81      else if (stack[top] == 'E' &&
```

Output

```
Enter expression (use i for identifier): i+i

Stack   Input   Action
$   i+i$    Shift
$i  +i$ Reduce E->i
$E  +i$ Shift
$E+ i$  Shift
$E+i    $   Reduce E->i
$E+E    $   Accept
```

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Run ▶

```
        else if (stack[top] == 'E' &&
                 (input[i] == '$' ||
                  input[i] == '+' ||
                  input[i] == '*'))
        {
            if (input[i] == '+' || input[i] == '*')
            {
                stack[++top] = input[i++];
                stack[top + 1] = '\0';
                printf("Shift\n");
            }
            else
            {
                printf("Reject\n");
                break;
            }
        }
        else
        {
            printf("Reject\n");
            break;
        }
    }

    return 0;
}

```

Output

```
Enter expression (use i for identifier): i+i

Stack   Input   Action
$   i+i$    Shift
$i  +i$ Reduce E->i
$E  +i$ Shift
$E+ i$  Shift
$E+i    $   Reduce E->i
$E+E    $   Accept
```